① a) $P(\{\underbrace{a}_{1}, \underbrace{b}_{2}, \underbrace{\{a,b\}}_{3}\}) \Rightarrow 2^n = 2^3 = 8$

b) $P(\{\underbrace{\emptyset}_{1}, \underbrace{a}_{2}, \underbrace{\{a\}}_{3}, \underbrace{\{\{a\}\}}_{4}\}) \Rightarrow 2^n = 2^4 = 16$

c) $P(\underbrace{P(\underbrace{\emptyset}_{0})}_{1}) \Rightarrow 2^0 = 1 \Rightarrow 2^1 = 2$

② A^n, $|A| = m$ elements | m^n

$n \geq 0$

③ a) $x \neq 2$
$x = -1$
$x = 0$
$x = 1$

$\{x \in \mathbb{Z} \mid x^2 < 3\} = \{-1, 0, 1\}$

b) $x \neq 0$
$x \neq 1$
$x = -1$

$\{x \in \mathbb{Z} \mid x^2 > x\} = \{\ldots, -3, -2, -1, 2, 3, \ldots\} = \mathbb{Z} - \{0, 1\}$

c) $x \neq 1$
$x \neq 0$
$x \neq -1$
$x = \frac{-1}{2}$

$\{x \in \mathbb{Z} \mid 2x+1 = 0\} = \emptyset$

5) a) $\overline{A\cap B\cap C} \subseteq \bar{A}\cup\bar{B}\cup\bar{C}$

$x\in(\overline{A\cap B\cap C})$

$x\notin(A\cap B\cap C)$

$x\notin A \vee x\notin B \vee x\notin C$

$x\notin(A\cup B\cup C)$

$x\in(\bar{A}\cup\bar{B}\cup\bar{C})$

$\bar{A}\cup\bar{B}\cup\bar{C} \subseteq \overline{A\cap B\cap C}$

$y\in(\bar{A}\cup\bar{B}\cup\bar{C})$

$y\notin(A\cup B\cup C)$

$y\notin A \vee y\notin B \vee y\notin C$

$y\notin(A\cap B\cap C)$

$y\in(\overline{A\cap B\cap C})$

b)

A	B	C	$\bar{A}$	$\bar{B}$	$\bar{C}$	$A\cap B\cap C$	$\overline{A\cap B\cap C}$	$= \bar{A}\cup\bar{B}\cup\bar{C}$
1	1	1	0	0	0	1	0	0
1	1	0	0	0	1	0	1	1
1	0	1	0	1	0	0	1	1
1	0	0	0	1	1	0	1	1
0	0	1	1	1	0	0	1	1
0	1	0	1	0	1	0	1	1
0	1	1	1	0	0	0	1	1
0	0	0	1	1	1	0	1	1

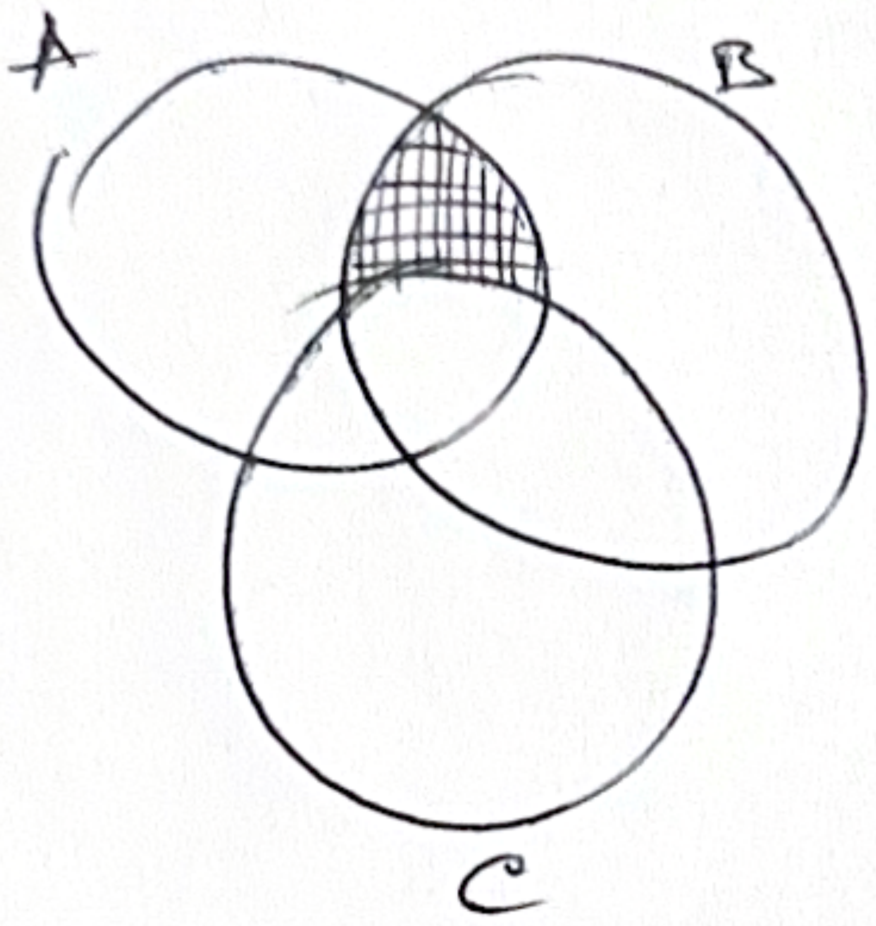
4) Ex!) $S=\{\overbrace{A,B,C}^{n}\}$

$2^n=2^3=8$

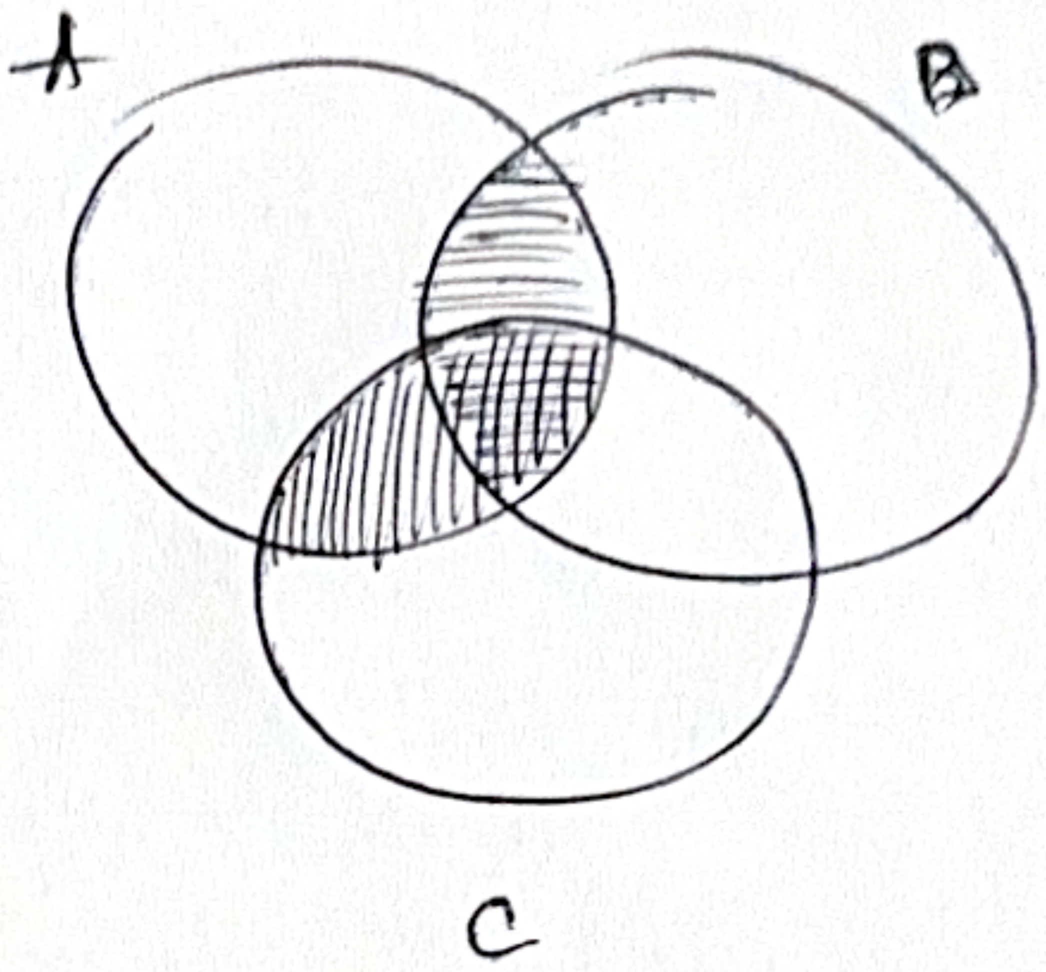
1	1	1	$\{A,B,C\}$
1	0	0	$\{A\}$
1	1	0	$\{A,B\}$
1	0	1	$\{A,C\}$
0	0	1	$\{C\}$
0	1	0	$\{B\}$
0	1	1	$\{B,C\}$
0	0	0	$\{\emptyset\}$

6

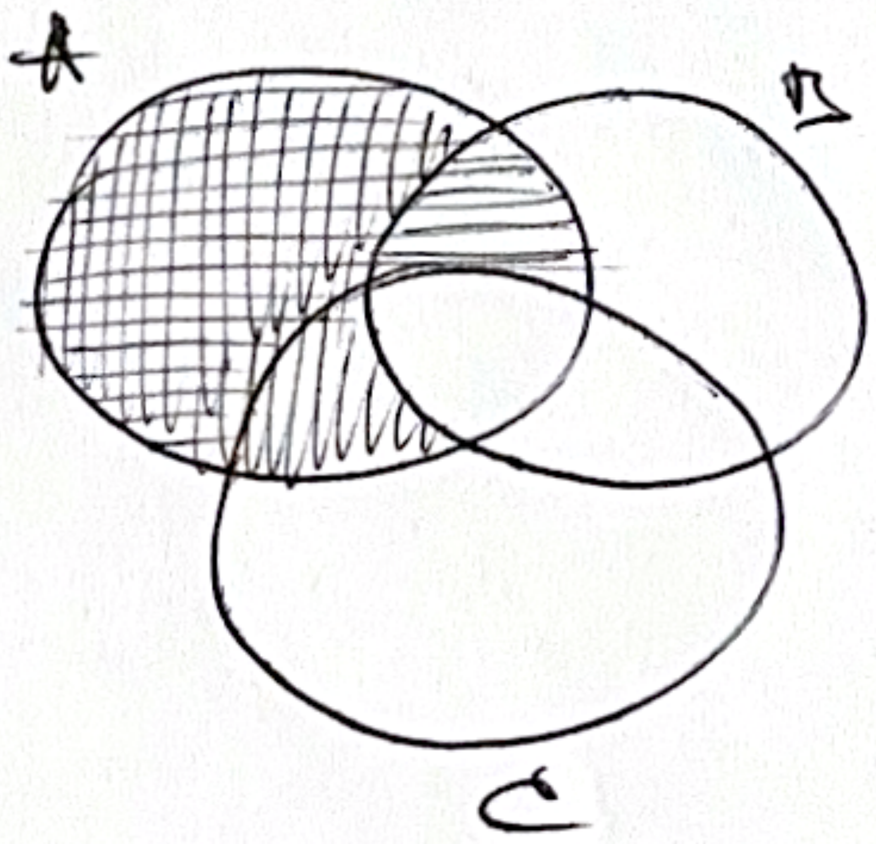
a)



b)



c)



7

a) Ex:) $A = \{1,2,3,4\}$, $B = \{3,4\}$

$A \cup B = \{1,2,3,4\} = A \Rightarrow B \subseteq A$

b) Ex:) $A = \{3,4\}$, $B = \{1,2,3,4\}$

$A \cap B = \{3,4\} = A \Rightarrow A \subseteq B$

c) Ex:) $A = \{1,2,3\}$, $B = \{4,5,6\}$

$A - B = \{1,2,3\} = A \Rightarrow A \cap B = \emptyset$

d) Ex:) $A = \{1,2,3\}$, $B = \{3,4,5\}$

$A \cap B = \{3\} = B \cap A \Rightarrow$ always holds

e) Ex:) $A = \{1,2,3\}$, $B = \{3,4,5\}$

$A - B = \{1,2\}$ and $B - A = \{4,5\}$

$A - B \neq B - A$, except $A = \emptyset, B = \emptyset \Rightarrow A - B = B - A$

⑧

$$A \oplus C = B \oplus C$$

$$(A \oplus C) \oplus C = (B \oplus C) \oplus C$$

$$A \oplus (C \oplus C) = B \oplus (C \oplus C)$$

$$A \oplus \emptyset = B \oplus \emptyset$$

$$A = B$$

⑨ a) $\{x \mid x \in \mathbb{Z} \wedge -i \leq x \leq i\}$

$A_1 = \{-1, 0, 1\}$

$A_2 = \{-2, -1, 0, 1, 2\}$

$A_3 = \{-3, -2, -1, 0, 1, 2, 3\}$

$A_i = \{-i, -i+1, \ldots, -1, 0, 1, \ldots i-1, i\}$

$\bigcup_{i=1}^{\infty} A_i = A_\infty = \mathbb{Z}$

$\bigcap_{i=1}^{\infty} A_i = A_1 = \{-1, 0, 1\}$

$A_1 \subset A_2 \subset A_3 \ldots\ldots$

b) $A_1 = \{-1, 1\}$

$A_2 = \{-2, 2\}$

$A_3 = \{-3, 3\}$

$A_1 \cup A_2 \cup_3 = \{-3, -2, -1, 1, 2, 3\}$

$\bigcup_{i=1}^{\infty} A_i = \mathbb{Z} - \{0\}$

$A_1 \cap A_2 \cap A_3 = \emptyset$

$\bigcap_{i=1}^{\infty} A_i = \emptyset$

c) $\{x \mid x \in \mathbb{R} \wedge -i \leq x \leq i\}$

$A_1 = \{-1; 1\}$

$A_2 = \{-2; 2\}$ $\quad \bigcup_{i=1}^{\infty} A_i = \mathbb{R}$

$A_3 = \{-3; 3\}$

$A_i = \{-i; i\}$

$$\begin{cases} A_1 \cap A_2 \cap A_3 = \{-1; 1\} \\ A_1 \subset A_2 \subset A_3 \ldots \\ \bigcap_{i=1}^{\infty} A_i = A_1 = \{x \mid x \in \mathbb{R} \wedge -1 \leq x \leq 1\} \\ = \{-1; 1\} \end{cases}$$

d) $\{x \mid x \in \mathbb{R} \wedge x \geq i\}$

$A_1 = \{1; \infty\}$

$A_2 = \{2; \infty\}$

$A_3 = \{3; \infty\}$

$A_i = \{i; \infty\}$

$$\begin{cases} A_1 \cap A_2 \cap A_3 = A_3 \\ A_1 \supset A_2 \supset A_3 \ldots A_\infty \\ \bigcap_{i=1}^{\infty} A_i = A_\infty = \{x \mid x \in \mathbb{R} \wedge x \geq \infty\} \\ = \emptyset \end{cases}$$

$\bigcup_{i=1}^{\infty} A_i = A_1 = \{1; \infty\} = \{x \mid x \in \mathbb{R} \wedge x \geq 1\}$